THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET C

Chapters 5–13: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity, Our Environment** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The opening of the stomach into the small intestine is regulated by: 1

(a) diaphragm **(b)** sphincter muscle **(c)** epiglottis **(d)** villi

2. The tiny pores on leaves through which gases are exchanged are called: 1

(a) lenticels **(b)** veins **(c)** root hairs **(d)** stomata

3. Which part of the brain controls involuntary actions like blood pressure and vomiting? 1

(a) pons **(b)** medulla **(c)** cerebellum **(d)** cerebrum

4. The gap between two neurons is called: 1

(a) impulse **(b)** dendrite **(c)** axon **(d)** synapse

5. For a unisexual flower, Student A gives papaya; Student B gives hibiscus. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The structure that becomes the embryo in a seed is formed from the: 1

(a) ovary wall **(b)** zygote **(c)** pollen grain **(d)** stigma

7. The number of chromosomes in a human gamete is: 1

(a) 46 **(b)** 23 **(c)** 22 **(d)** 44

8. The phenotype of RrYy is: 1

(a) wrinkled green **(b)** round green **(c)** round yellow **(d)** wrinkled yellow

9. In some reptiles, the sex of the offspring is determined by: 1

(a) the mother only

(b) X and Y chromosomes

(c) food

(d) the temperature at which eggs are incubated

10. A cross between Tt and Tt gives the proportion of tall plants as: 1

(a) 50% **(b)** 25% **(c)** 75% **(d)** 100%

11. The organisms at the first trophic level are: 1

(a) carnivores **(b)** herbivores **(c)** producers **(d)** decomposers

12. Ozone is a molecule made of: 1

(a) three oxygen atoms **(b)** two oxygen atoms

(c) one oxygen atom **(d)** oxygen and carbon

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Why are the walls of ventricles thicker than those of atria? 2

14. A student says that movement of the sensitive plant on touch is a growth movement. Justify or refute. 2

15. What is the importance of variation for a species? 2

16. A student says sex chromosomes in females are dissimilar. Justify or refute. 2

17. What are trophic levels? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. What is translocation? Why is it essential for plants? How does it occur? 3

19. Why is the brain called the main coordinating centre? Name the three major parts. 3

20. What is the importance of DNA copying in reproduction? Why are copies not identical? 3

21. What are the advantages of reusable over disposable items (three)? 3

22. Draw a labelled diagram of a nephron (describe its parts) and state how urine is formed. 3

OR

What is the role of acid in the stomach and of the liver in digestion? 3

23. Explain the role of adrenaline in an emergency situation. 3

OR

Differentiate between the nervous and hormonal mechanisms of coordination (three points). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** A student places a potted plant in a dark room with light coming from one window. After a few days the shoot bends towards the window. 5

(i) Name the phenomenon.
(ii) Name the hormone involved.
(iii) Explain the mechanism.
(iv) If the plant is turned around, what happens?

OR

Case study: A child is unusually short; another person has excessive thirst and urination with high blood sugar. 5

(i) Which gland is affected in the first case and which hormone?
(ii) Name the disorder in the second case.
(iii) Name the hormone and gland involved.
(iv) Suggest one treatment for the second case.

25. **Case study:** A couple has two daughters. Relatives blame the mother for not having a son. 5

(i) Who determines the sex of the child?
(ii) Explain with a flow chart.
(iii) What is the probability of a boy in each pregnancy?
(iv) How should the couple respond?

OR

(i) Explain Mendel's monohybrid cross with a checkerboard (Punnett square). 5
(ii) State the law of dominance.
(iii) What do you mean by F1 and F2?

END OF QUESTION PAPER — Total: 50 Marks